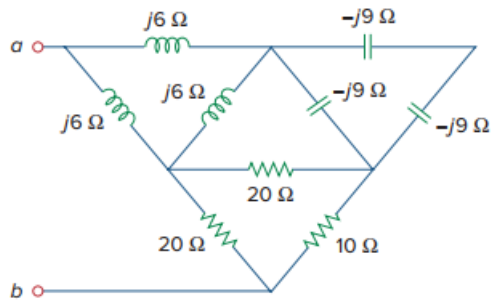


Problem

Calculate the value of Z_{ab} in the network of Fig. 9.79.

Figure 9.79:

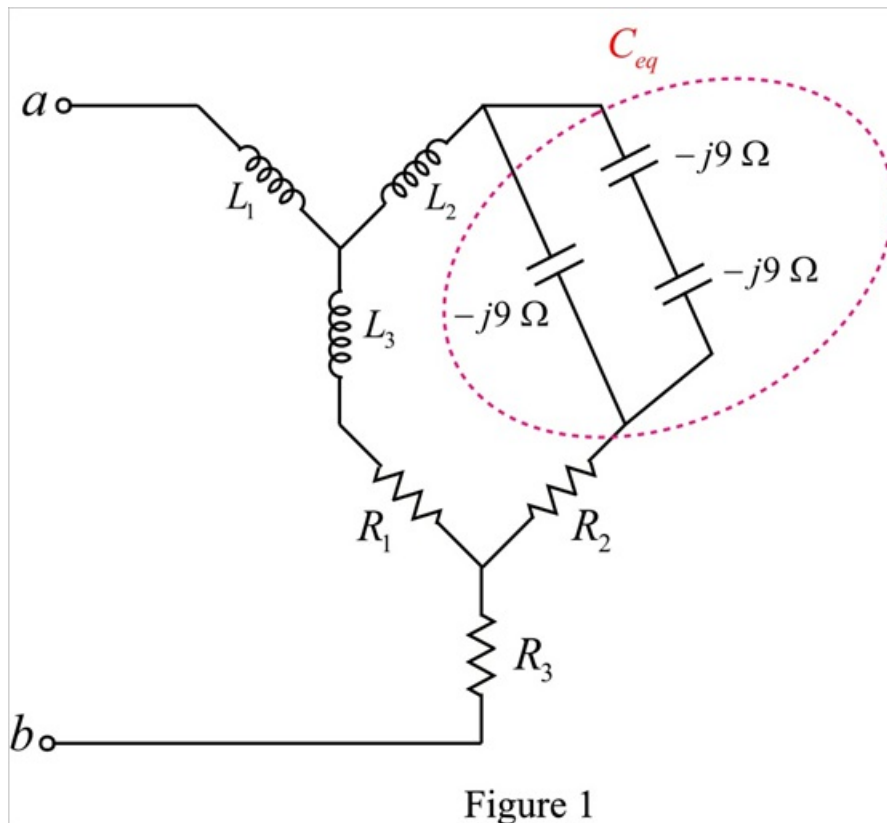


Step-by-step solution

Step 1 of 6

Refer to figure 9.79 for the circuit given in the textbook.

Transform the delta networks of inductors and resistors as shown in figure 1.



Step 2 of 6

Calculate the equivalent impedance C_{eq} .

$$\begin{aligned}C_{eq} &= (-j9 - j9) \parallel (-j9) \\&= (-j18) \parallel (-j9) \\&= \frac{(-j18)(-j9)}{-j18 - j9} \\&= \frac{(-j18)(-j9)}{(-j9)(3)} \\&= -j6 \Omega\end{aligned}$$

Step 3 of 6

Calculate the impedance of inductance L_1 .

$$\begin{aligned}L_1 &= \frac{(j6)(j6)}{(j6) + (j6) + (j6)} \\&= \frac{(j6)(j6)}{3 \times (j6)} \\&= j2 \Omega\end{aligned}$$

Calculate the impedance of inductance L_2 .

$$\begin{aligned}L_2 &= \frac{(j6)(j6)}{(j6) + (j6) + (j6)} \\&= \frac{(j6)(j6)}{3 \times (j6)} \\&= j2 \Omega\end{aligned}$$

Calculate the impedance of inductance L_3 .

$$\begin{aligned}L_3 &= \frac{(j6)(j6)}{(j6) + (j6) + (j6)} \\&= \frac{(j6)(j6)}{3 \times (j6)} \\&= j2 \Omega\end{aligned}$$

Step 4 of 6

Calculate the resistance R_1 .

$$\begin{aligned} R_1 &= \frac{20 \times 20}{20 + 20 + 10} \\ &= \frac{400}{50} \\ &= 8 \, \Omega \end{aligned}$$

Calculate the resistance R_2 .

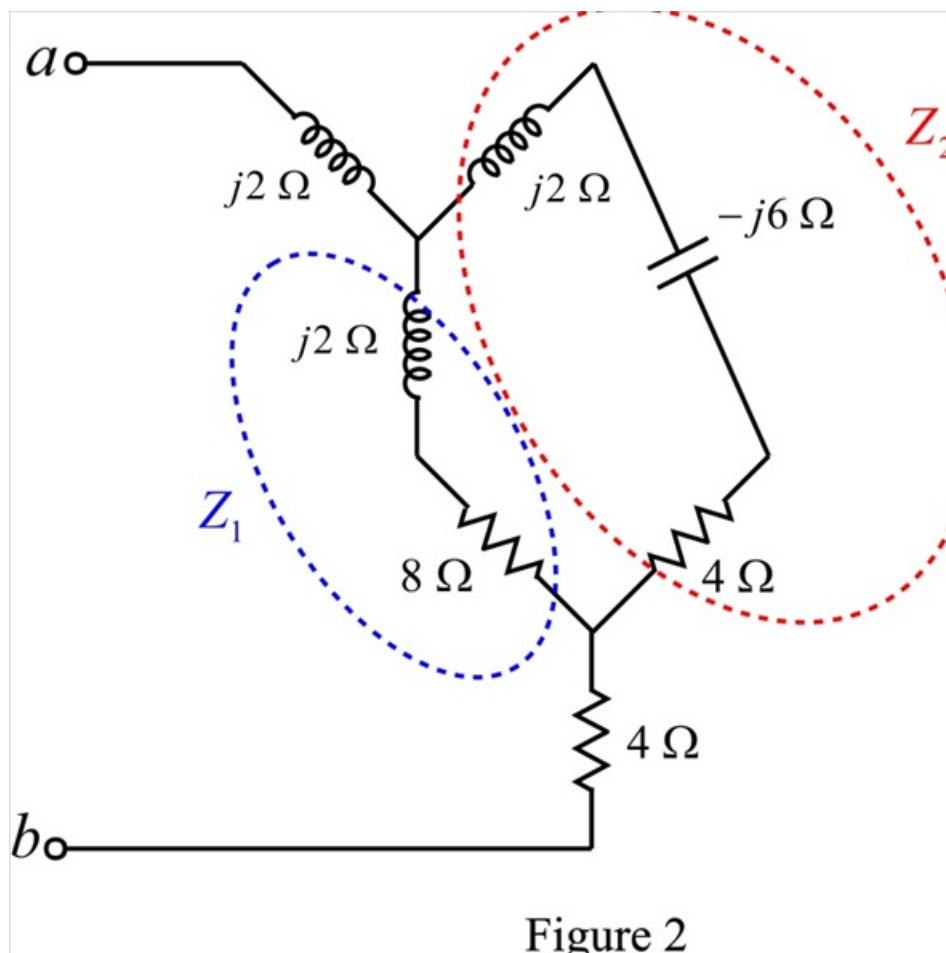
$$\begin{aligned} R_2 &= \frac{10 \times 20}{20 + 20 + 10} \\ &= \frac{200}{50} \\ &= 4 \, \Omega \end{aligned}$$

Calculate the resistance R_3 .

$$\begin{aligned} R_3 &= \frac{10 \times 20}{20 + 20 + 10} \\ &= \frac{200}{50} \\ &= 4 \, \Omega \end{aligned}$$

Step 5 of 6

Redraw the circuit as shown in figure 2.



Step 6 of 6

Calculate the impedance Z_1 .

$$Z_1 = (8 + j2) \Omega$$

Calculate the impedance Z_2 .

$$\begin{aligned} Z_2 &= 4 + j2 - j6 \\ &= (4 - j4) \Omega \end{aligned}$$

Calculate the impedance across the terminals a and b .

$$\begin{aligned} Z_{ab} &= j2 + (Z_1 \parallel Z_2) + 4 \\ &= j2 + \left(\frac{Z_1 Z_2}{Z_1 + Z_2} \right) + 4 \end{aligned}$$

Substitute $8 + j2$ for Z_1 , and $4 - j4$ for Z_2 .

$$\begin{aligned} Z_{ab} &= j2 + \left(\frac{(8 + j2)(4 - j4)}{(8 + j2) + (4 - j4)} \right) + 4 \\ &= j2 + \left(\frac{32 - j32 + j8 + 8}{12 - j2} \right) + 4 \\ &= j2 + \frac{40 - j24}{12 - j2} + 4 \\ &= (7.567 + j0.5946) \Omega \end{aligned}$$

Therefore, the impedance Z_{ab} is $\boxed{(7.567 + j0.5946) \Omega}$.